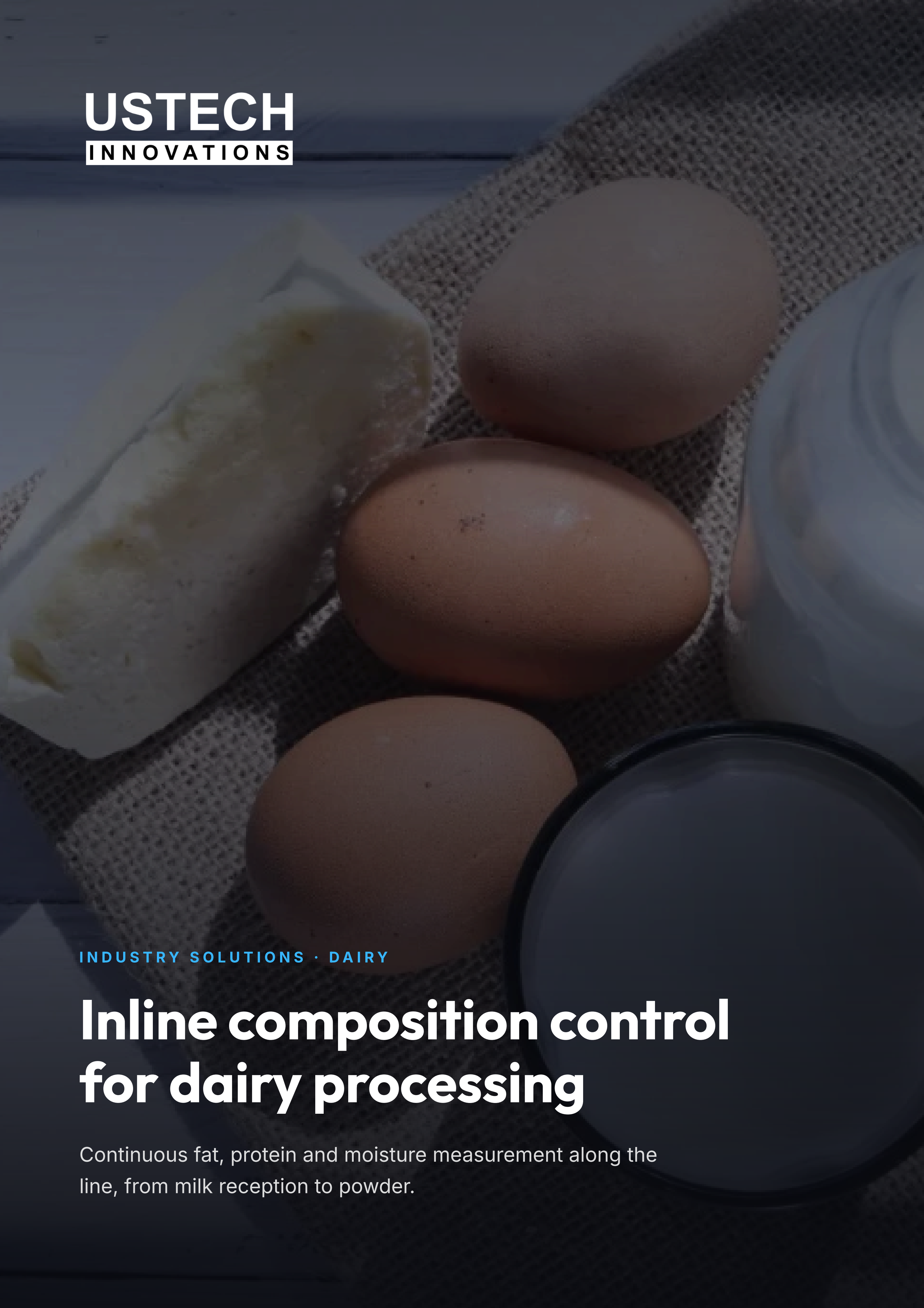

USTECH
INNOVATIONS

INDUSTRY SOLUTIONS · DAIRY

Inline composition control for dairy processing

Continuous fat, protein and moisture measurement along the line, from milk reception to powder.

Where measurement supports your process

Four points along the production flow where NIR results replace waiting for the laboratory.

01

Milk reception

Inline measurement at the receiving dock verifies the composition of each incoming load.

Parameters	Fat, protein, lactose, total solids
Instruments	ProLine17EC · ProLine17ECS · ProLine2550

02

Separation and fat standardization

Measurement downstream of the separator feeds the standardization PLC.

Parameters	Fat, protein, total solids
Instruments	ProLine17EC · ProLine17ECS · ProLine2550 + ProChem

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03

Cheese, yogurt and whey

Monitoring of whey lines and fermented product lines for consistent composition.

Parameters	Residual fat in whey, total solids, protein
Instruments	ProLine17EC · ProLine17ECS · ProLine2550

04

Drying and powder

Moisture tracking at the dryer outlet keeps powder close to the target moisture.

Parameters	Moisture, protein, fat
Instruments	ProLine17EC · ProLine17ECS · ProLine2550 + caliX · TrueLine 17EC for spot checks at the sampling point

Instruments and software

ProLine17EC

Inline NIR analyzer

- 950–1650 nm, InGaAs photodiode array
- Typical analysis time 2–20 s
- 316 stainless steel, sapphire window, IP65
- Solid and liquid samples, no bypass

ProLine17ECS

Inline NIR analyzer

- 950–1650 nm, InGaAs photodiode array
- 2000 spectra per sample, S/N 23,000:1 (typical)
- 316 stainless steel, sapphire window, IP65
- Typical analysis time 2–20 s

ProLine
2550

ProLine2550

Inline FT-NIR analyzer

- 1,350–2,550 nm FT-NIR
- HTTP interface to the plant PLC (Ethernet, TCP/IP)
- IP65, -5 to +45 °C

TrueLine 17EC

Handheld NIR analyzer

- 908–1676 nm
- Measures at the sampling point
- Mobile app and cloud portal, Bluetooth
- Weight 1.4 kg

ProChem

Instrument control and results. Shows measurements in real time, exports reports, and passes results to the plant PLC over an Ethernet-based HTTP interface.

caliX

Calibration workspace. Build, validate and transfer calibration models, with automated preprocessing and model selection.

Start with your application

Five answers are enough for us to tell you which instrument fits and where it should be installed.

1. What will be measured?
   Material and form: raw milk, cream, skim milk, whey, curd, milk powder.
2. Which parameters do you need?
   For example fat, protein, lactose, total solids, moisture.
3. Where should the measurement take place?
   Laboratory, next to the line, or directly on a pipe or at the dryer outlet.
4. How do you measure these values today?
   Reference method and how long a result takes.
5. Where is the plant and what is your timeframe?
   Country and planned start.

Contact us

USTECH Innovations LLC
600 Park Offices Dr, Ste 300
Durham, NC 27713, USA

info@ustechinnovations.com
www.ustechinnovations.com